

OSSP:

Commands:

Uname : It displays the System information

Lscpu : It displays the complete CPU information

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root@LAPTOP-4H877K4U:~# nano process.c
root@LAPTOP-4H877K4U:~# gcc process.c -o process
root@LAPTOP-4H877K4U:~# ./process
Enter Linux command: lscpu

Child Process
Child PID = 835
Architecture:           x86_64
CPU op-mode(s):         32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                  16
On-line CPU(s) list:    0-15
Vendor ID:               GenuineIntel
Model name:              13th Gen Intel(R) Core(TM) i7-13620H
CPU family:              6
Model:                   186
Thread(s) per core:      2
Core(s) per socket:      8
Socket(s):               1
Stepping:                2

```

Ps : It displays the running process of our file

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root@LAPTOP-4H877K4U:~# nano process.c
root@LAPTOP-4H877K4U:~# gcc process.c -o process
root@LAPTOP-4H877K4U:~# ./process
Enter Linux command: ps

Child Process
Child PID = 910
  PID TTY          TIME CMD
  755 pts/2        00:00:00 bash
  842 pts/2        00:00:00 process
  910 pts/2        00:00:00 ps

Parent Process
Parent PID = 842
Child PID = 910

```

Relation Between Hardware Resources and Operating System Services:

1. CPU

The operating system shares the CPU among multiple programs by giving each one a turn to execute.

Example: When we open Chrome and a music player together, the OS schedules CPU time for both. It Process Both Chrome And music player

2. Memory (RAM) -> Memory Management

E.Pranay ReddyThe operating system gives memory to each program that is running It makes sure one program does not use another program's memory. Example: Chrome and VS Code each get their own memory space while running.

3. Storage (SSD) → File System Management

The operating system storesand organizes files on the storage device.
Example: Saving a document in the Documents folder on our computer.

4. I/O Devices → Device Management

The operating system uses device drivers to control input and output devices.